Mohammed Rashid Chowdhury

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PROFESSIONAL SUMMARY

Senior Engineer with 7+ years of experience specializing in Machine Learning and Large-Scale Data Engineering. Proven track record of architecting GenAI/RAG pipelines, optimizing petabyte-scale Spark workloads, and deploying cloud-native AI products. Expert in bridging the gap between complex AI research and production-ready software solutions for enterprise clients.

EXPERIENCE

Senior Machine Learning & Data Engineer — May 2022 – Present

CGI Inc. · *Toronto, ON (Remote)*

- **Advanced AI Product Development:** Engineered high-performance RAG (Retrieval-Augmented Generation) pipelines by integrating LangChain with Neo4j knowledge graphs and FAISS vector stores, enabling context-aware retrieval and improving user engagement by 50%.
- **Intelligent Data Deduplication & Reconciliation:** Designed and deployed a BERT-based entity matching solution using TensorFlow to automate data deduplication and reconciliation across disparate enterprise datasets.
- **Internal AI Strategy & Adoption:** Partnered directly with cross-functional teams to define architecture standards and cost-efficient POCs for cloud-native ML services, facilitating the adoption of AI tools across the organization.
- **Distributed Data Engineering:** Optimized large-scale Apache Spark workloads and Delta Lake tables, reducing query latency and operational overhead by 40%.
- **AI Reliability & Compliance:** Orchestrated data lineage and PII anonymization using Databricks Unity Catalog to ensure regulatory compliance and high-fidelity data for AI experimentation.

Machine Learning / Data Engineer II (MLOps Team) — July 2021 – May 2022

Bell Canada · *Toronto, ON*

- **Real-Time AI Infrastructure:** Engineered event-driven pipelines using Apache Kafka and Spark Structured Streaming to process high-velocity telemetry data with sub-second latency, enabling automated model re-training and real-time inference.
- **Production API Engineering:** Deployed ML models as production-ready, high-throughput API endpoints using BentoML and FastAPI, ensuring scalable delivery of model predictions to downstream microservices.
- **Model Observability & Integrity:** Developed a drift detection framework using Prometheus, Grafana, and Python to monitor statistical deviations (KL-divergence/KS-test) in production data, ensuring the reliability of AI outcomes.
- **Cloud-Native Deployment:** Orchestrated Jenkins CI/CD pipelines for containerized workloads on Kubernetes/OpenShift; utilized multi-stage Docker builds to reduce image footprints by 50%, accelerating deployment cycles and reducing storage overhead.
- **Automated Quality Gates:** Designed data quality management processes to detect anomalies in source datasets, preventing corrupted data from entering the training lifecycle.

Software Engineer (Machine Learning Framework) — February 2019 – July 2021

Siemens Digital Industries Software · *Saskatoon, SK*

- **Performance Engineering:** Accelerated application start times by 2x using cProfile for bottleneck identification and implementing high-performance caching with Redis and Python generators.
- **Algorithmic Development:** Implemented proprietary ML algorithms for chip verification using Python, C++, and Cython to enhance computational efficiency.
- **Cloud Infrastructure:** Collaborated with DevOps to streamline CI/CD using AWS (ECR, ECS, CodePipeline) and Ansible for scalable tool deployment.

Graduate Research Assistant (Parallel and Distributed Programming) — September 2016 – February 2019

DISCUS lab, University of Saskatchewan · *Saskatoon, SK*

- **Distributed Computing:** Scaled a CNN-based flower counting application by ∼7X on multi-GPU clusters using TensorFlow (Parameter Server), Horovod (Ring Reduce), and OpenMPI (communication).
- **System Performance:** Accelerated image registration tasks by ∼3x using CUDA C++ and Cython, implementing custom resource scheduling for efficient GPU utilization.

- **Big Data Processing:** Developed distributed data processing pipelines using Apache Spark, HDFS, and Hive to manage and analyze high-volume image datasets.

TECHNICAL SKILLS

AI & GenAI: Large Language Models (LLMs), RAG Pipelines, LangChain, FAISS, Embeddings, AI Agents, TensorFlow, BERT.

Data Engineering: Apache Spark (PySpark), Kafka, Delta Lake, Hive, Airflow, Hadoop, ETL/ELT.

Cloud & DevOps: GCP (Vertex AI, GCS), Azure, AWS, Docker, Kubernetes (GKE/OpenShift), CI/CD (Jenkins, Azure DevOps), Terraform.

Languages & Tools: Python (Expert), SQL, Bash, C++, FastAPI, MLflow, Prometheus, Grafana.

EDUCATION

- **M.Sc. in Computer Science** – University of Saskatchewan (Sep 2016 – Feb 2019)
- **B.Sc. in Computer Science and Engineering** – North South University, Dhaka, Bangladesh (May 2009 – Sep 2015)

CERTIFICATIONS

- **Certified Azure Data Engineer Associate**, Microsoft (2024 – 2025)

AWARDS

- **CGI Impact Award (NCR F23 Q2)** – Awarded for exceeding expectations and making significant contributions that benefit CGI Stakeholders (GC - NextGen).
- **Winner of Emerging Agriculture Hackathon 2017** – First place for building an image processing application identifying crop pest infestations and plant nutrient deficiencies. (Media coverage link)
- **Faculty Scholarship Award** – Awarded for academic achievement covering full cost of Master's Degree program at U of S.